

## FOLKARNA, Sweden

WMO: 024440

Lat: 60.167N

Lon: 16.317E

Elev: 230

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 82-94

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.0       | -2.6 | -12.6                         | 2.8 | -8.7 | -6.1 | 4.0 | -2.3 | 34.3                  | 33.4 | 27.4 | 33.3 | 2.2                   | 20   | 0.596 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.0                   | 80.4            | 63.6 | 76.9 | 61.8 | 73.2 | 59.9 | 65.2                | 76.4 | 63.5 | 73.4 | 61.6 | 70.6 | 5.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 61.6                            | 82.5 | 68.2 | 59.3 | 75.9 | 66.9 | 57.3 | 70.8 | 65.1 | 30.2          | 75.4 | 28.9 | 73.2 | 27.6 | 70.9 | 70.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 22.1              | 16.1 | 12.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 41.2   | 23.7  | 23.6  | 28.8  | 38.6  | 49.9  | 56.4  | 61.4  | 58.4  | 49.3  | 43.0  | 32.6  | 26.8 |      |
|                                                                                  | DBStd                                                                            | 15.51  | 12.62 | 11.74 | 9.07  | 6.88  | 6.23  | 6.22  | 4.99  | 5.18  | 5.20  | 5.91  | 7.73  | 8.95 |      |
|                                                                                  | HDD50                                                                            | 4196   | 815   | 738   | 659   | 350   | 79    | 14    | 0     | 3     | 72    | 226   | 522   | 718  |      |
|                                                                                  | HDD65                                                                            | 8745   | 1280  | 1158  | 1123  | 793   | 468   | 265   | 135   | 216   | 470   | 682   | 972   | 1183 |      |
|                                                                                  | CDD50                                                                            | 967    | 0     | 0     | 0     | 7     | 76    | 206   | 354   | 262   | 53    | 9     | 0     | 0    |      |
|                                                                                  | CDD65                                                                            | 41     | 0     | 0     | 0     | 0     | 0     | 7     | 24    | 10    | 0     | 0     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 650    | 0     | 0     | 0     | 3     | 40    | 133   | 351   | 120   | 3     | 0     | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 118    | 0     | 0     | 0     | 0     | 1     | 17    | 79    | 21    | 0     | 0     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 4.8    | 5.6   | 4.9   | 4.9   | 5.1   | 5.5   | 4.6   | 4.2   | 4.0   | 4.4   | 4.5   | 4.5   | 5.2  |      |
| Precipitation                                                                    | PrecAvg                                                                          | 23.7   | 1.6   | 1.2   | 1.2   | 1.3   | 1.7   | 2.5   | 2.9   | 2.8   | 2.2   | 2.2   | 2.2   | 1.9  |      |
|                                                                                  | PrecMax                                                                          | 30.0   | 3.1   | 2.6   | 2.4   | 3.6   | 3.0   | 4.8   | 6.3   | 6.2   | 5.4   | 5.5   | 5.0   | 3.4  |      |
|                                                                                  | PrecMin                                                                          | 19.8   | 0.3   | 0.1   | 0.2   | 0.2   | 0.5   | 1.1   | 0.6   | 0.9   | 0.2   | 0.7   | 0.9   | 0.4  |      |
|                                                                                  | PrecStd                                                                          | 2.6    | 0.8   | 0.6   | 0.6   | 0.8   | 0.7   | 1.1   | 1.3   | 1.3   | 1.3   | 1.2   | 1.1   | 0.8  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 46.2  | 48.7  | 54.9  | 69.6  | 78.6  | 82.5  | 87.4  | 83.2  | 70.2  | 60.9  | 51.1  | 46.1 |      |
|                                                                                  |                                                                                  | MCWB   | 41.2  | 43.5  | 45.7  | 56.7  | 60.8  | 64.7  | 67.5  | 65.1  | 58.2  | 54.4  | 47.2  | 41.9 |      |
|                                                                                  | 2%                                                                               | DB     | 43.0  | 45.6  | 49.0  | 61.5  | 73.6  | 77.6  | 81.4  | 77.3  | 64.5  | 56.4  | 48.5  | 43.0 |      |
|                                                                                  |                                                                                  | MCWB   | 38.6  | 41.2  | 42.4  | 49.0  | 57.7  | 62.1  | 64.3  | 62.8  | 55.3  | 52.0  | 44.4  | 40.4 |      |
|                                                                                  | 5%                                                                               | DB     | 40.4  | 40.9  | 43.7  | 55.3  | 67.9  | 74.1  | 77.8  | 72.0  | 61.4  | 53.7  | 45.7  | 40.1 |      |
|                                                                                  |                                                                                  | MCWB   | 36.3  | 37.0  | 37.9  | 45.5  | 54.8  | 60.2  | 62.6  | 60.1  | 53.5  | 49.7  | 42.7  | 38.1 |      |
|                                                                                  | 10%                                                                              | DB     | 38.1  | 37.6  | 39.7  | 50.2  | 63.6  | 70.2  | 73.8  | 68.8  | 58.5  | 51.7  | 43.0  | 37.6 |      |
|                                                                                  |                                                                                  | MCWB   | 35.2  | 35.4  | 35.2  | 42.9  | 52.1  | 57.8  | 60.8  | 58.4  | 51.8  | 48.5  | 41.0  | 35.8 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 42.2  | 44.5  | 46.8  | 56.2  | 61.2  | 65.5  | 68.9  | 66.5  | 61.2  | 56.5  | 48.0 | 42.9 |
|                                                                                  |                                                                                  |        | MCDB  | 45.4  | 48.2  | 53.2  | 69.7  | 76.0  | 78.4  | 82.9  | 78.2  | 67.5  | 58.3  | 50.2 | 44.9 |
| 2%                                                                               |                                                                                  | WB     | 39.4  | 41.7  | 42.9  | 49.5  | 58.2  | 63.6  | 66.2  | 64.2  | 57.5  | 53.0  | 44.9  | 40.6 |      |
|                                                                                  |                                                                                  | MCDB   | 42.2  | 45.0  | 48.8  | 58.2  | 70.5  | 75.2  | 77.7  | 73.8  | 62.8  | 55.6  | 47.1  | 42.9 |      |
| 5%                                                                               |                                                                                  | WB     | 37.3  | 38.1  | 38.3  | 46.4  | 55.8  | 61.5  | 64.2  | 62.0  | 55.0  | 50.9  | 43.0  | 38.3 |      |
|                                                                                  |                                                                                  | MCDB   | 39.4  | 40.4  | 42.9  | 53.6  | 65.8  | 72.0  | 74.4  | 69.5  | 59.2  | 52.9  | 45.4  | 39.8 |      |
| 10%                                                                              |                                                                                  | WB     | 35.5  | 35.5  | 35.8  | 43.9  | 53.2  | 59.0  | 62.2  | 60.1  | 53.2  | 48.7  | 41.1  | 36.1 |      |
|                                                                                  |                                                                                  | MCDB   | 37.9  | 37.2  | 38.7  | 49.3  | 61.9  | 68.2  | 71.3  | 67.3  | 57.0  | 50.9  | 42.9  | 37.3 |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 10.9  | 14.0  | 14.7  | 16.0  | 19.5  | 18.1  | 19.0  | 17.1  | 15.4  | 11.4  | 8.8  | 10.2 |
|                                                                                  |                                                                                  |        | MCDBR | 10.4  | 12.3  | 17.8  | 26.0  | 28.6  | 25.8  | 27.8  | 23.9  | 19.6  | 13.6  | 10.3 | 11.2 |
|                                                                                  | MCWBR                                                                            |        | 9.1   | 9.8   | 12.0  | 15.1  | 14.8  | 12.1  | 12.1  | 10.9  | 11.2  | 9.5   | 8.6   | 10.1 |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 9.7   | 10.1  | 16.0  | 22.4  | 25.3  | 23.8  | 23.1  | 20.9  | 16.4  | 10.6  | 8.9   | 10.6 |      |
|                                                                                  |                                                                                  | MCWBR  | 9.2   | 8.5   | 11.1  | 13.7  | 14.1  | 12.0  | 11.2  | 10.4  | 10.4  | 8.7   | 7.9   | 10.1 |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.261  | 0.307 | 0.330 | 0.366 | 0.364 | 0.356 | 0.378 | 0.373 | 0.337 | 0.324 | 0.287 | 0.242 |      |      |
|                                                                                  | taud                                                                             | 2.201  | 2.290 | 2.362 | 2.345 | 2.409 | 2.454 | 2.410 | 2.420 | 2.509 | 2.445 | 2.264 | 2.112 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 164    | 220   | 254   | 264   | 274   | 278   | 267   | 259   | 252   | 214   | 151   | 119   |      |      |
|                                                                                  | Edh at Noon                                                                      | 13     | 21    | 27    | 33    | 33    | 32    | 33    | 30    | 23    | 18    | 12    | 9     |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 91     | 294   | 769   | 1230  | 1571  | 1753  | 1617  | 1269  | 830   | 374   | 116   | 57    |      |      |
|                                                                                  | RadStd                                                                           | 14     | 49    | 93    | 125   | 156   | 120   | 168   | 132   | 103   | 62    | 28    | 6     |      |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (54 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -207        | N/A   | N/A   | N/A   |

Nomenclature: See separate page